

Session 6: Bayesian Nonparametric Methods

Modern Macroeconometrics

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Day 3 (morning session), 9 July 2026



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OVERVIEW OF SESSION 6

- **Motivation:** Flexible, data-driven functional forms via nonparametric approaches
- **Bayesian additive regression trees (BART):** Nonlinearity via a sum of shallow decision trees; strong regularisation priors; Bayesian backfitting
- **Gaussian processes (GPs):** Nonlinearity via a prior over smooth functions; so-called kernels encode structure
- **Bayesian neural networks (BNNs):** Universal approximators, nesting linear regression and approximating BART and GP

Nonparametric regressions

MOTIVATION FOR NONPARAMETRIC APPROACHES

- Economic relationships change over time for a variety of reasons
- Standard econometric models assume **stability of parameters**
- Parametric approaches for overcoming this limitation
 - Regime-switching models
 - Interacted models
 - Time-varying parameter models (Session 3)
- **Shortcomings** of these **parametric models**
 - They depend on specific assumptions on parameter evolution (abrupt breaks, gradual drift)
 - An improvement over linear models, but still prone to misspecification
- **Nonparametric methods** provide a flexible alternative

NONPARAMETRIC APPROACHES IN MACROECONOMICS

- Nonparametric approaches are a **powerful tool**
- The combination of nonparametric methods with multivariate time series models has become increasingly popular
- **Agnostic on the functional relationships** between the endogenous variables and regressors (e.g., their lags)
- In a structural VAR, they allow for the analysis of potential **asymmetries in transmission channels** following an exogenous shock
 - For example, positive monetary policy shocks (tightening) may have different effects on the macroeconomy than negative monetary shocks (easing)
 - A priori, we do not know the exact nature of these effects, but we aim to allow for this flexibility, estimate it from the data, and let the model “speak” if there are indeed asymmetries

FUNCTION-SPACE VIEW OF A REGRESSION

- A generalised regression relates y_t to regressors x_t :

$$y_t = f(x_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- f is an **unknown function** mapping the input x_t to the output y_t
- Rather than estimating a fixed set of parameters, we **look for the “best” function** describing the relationship between x_t and y_t
- Nonlinear transformations of x_t are called **features**, expanding the input into a higher-dimensional **feature space** $\phi(x_t)$
- For example: $\phi(x_t) = (x_t, x_t^2, x_t^3)'$

FUNCTION-SPACE VIEW OF A REGRESSION (CONT.)

- › f is typically assumed **linear**: $f(x_t) = \beta x_t$; a pretty **strong assumption**
- › A **polynomial regression** extends the linear model to nonlinearities

$$f(\phi(x_t)) = \beta_1 x_t + \beta_2 x_t^2 + \beta_3 x_t^3$$

- › Still **linear in parameters** and even OLS estimation is possible
- › But this is a **deterministic** projection of x_t onto a feature space $\phi(x_t)$
- › **Infinitely many** such mappings exist
- › Choosing the “right” features is hard; nonparametric methods let the **data** decide

Bayesian additive regression trees (BART)

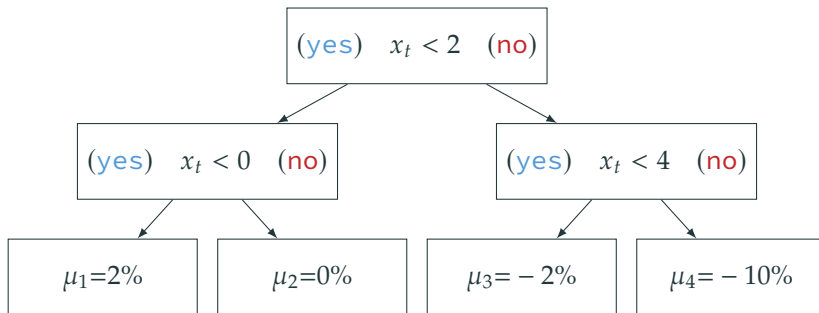
A REGRESSION TREE MODEL

- › A simple regression tree is given by

$$y_t = g(\mathbf{x}_t \mid \mathcal{T}, \boldsymbol{\mu}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- › g is a tree function determined by $\mathcal{T}$, the **tree structure**
- › $\boldsymbol{\mu} = (\mu_1, \dots, \mu_b)'$ is the vector of **terminal node parameters**
- › $\mathcal{T}$ collects decision rules $\{x_t < c\}$ or $\{x_t \geq c\}$, with x_t the **splitting variable** and c a threshold
- › At each **node** the algorithm decides whether it is terminal or whether to split further (the tree grows); once all nodes are terminal we have $\mathcal{T}$ of depth D
- › Each of the b **terminal nodes** ($j = 1, \dots, b$) carries a mean μ_j

A SIMPLE GRAPHICAL REPRESENTATION OF A REGRESSION TREE



- For example, GDP growth (y_t) on lagged NFCI ($x_t = \text{NFCI}_{t-1}$): GDP growth may be more reactive to NFCI once a certain threshold is crossed

ILLUSTRATION OF A REGRESSION TREE

› Equivalently, using dummy variables:

› if $x_t < 2$: $D_{00,t} = 1$, else 0

› if $x_t < 0$: $D_{11,t} = 1$, else 0

› if $x_t < 4$: $D_{12,t} = 1$, else 0

$$y_t = \begin{cases} \mu_1 + \varepsilon_t, & \text{if } D_{00,t} = 1 \text{ \& } D_{11,t} = 1 \\ \mu_2 + \varepsilon_t, & \text{if } D_{00,t} = 1 \text{ \& } D_{11,t} = 0 \\ \mu_3 + \varepsilon_t, & \text{if } D_{00,t} = 0 \text{ \& } D_{12,t} = 1 \\ \mu_4 + \varepsilon_t, & \text{if } D_{00,t} = 0 \text{ \& } D_{12,t} = 0 \end{cases} \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

› Trees are **nonparametric**: Number of parameters is not fixed a priori; it becomes known only after estimation

AN ADDITIVE REGRESSION TREE MODEL

- › BART (Chipman, George & McCulloch 2010, *AoAS*) approximates the unknown f as a **sum of S trees**:

$$f(x_t) \approx \sum_{s=1}^S g(x_t \mid \mathcal{T}_s, \mu_s)$$

- › If the trees were unrestricted, this would give **perfect fit**
- › Instead, most trees take a very **simple form**, with each being a “**weak learner**”, explaining only a little of the variation in y_t
- › **Summing over many trees** yields an additive model flexible enough to capture the dynamics of y_t ; and under a suitable prior, one that does not overfit

BART PRIORS

- ▶ Priors on the trees $\mathcal{T}_s$, the leaf values μ_s , and the error variance σ^2 (inverse Gamma, as in Session 2)
- ▶ **Tree prior** keeps trees **shallow**: A node at depth η splits with probability $\alpha/(1 + \eta)^\beta$; the default $\alpha = 0.95$, $\beta = 2$ avoids growing trees too deep
- ▶ **Leaf prior** $\mu_j \sim \mathcal{N}(0, \sigma_\mu^2)$, with σ_μ shrinking as the number of trees S grows
- ▶ Splitting variables are drawn uniformly over the K predictors

THE TREE PRIOR IN NUMBERS ($\alpha = 0.95, \beta = 2$)

A node at depth η splits with probability $p_{\text{split}}(\eta) = \frac{0.95}{(1 + \eta)^2}$

depth η	0	1	2	3	4
$p_{\text{split}}(\eta)$	0.95	0.24	0.11	0.06	0.04
Pr(a branch reaches depth η)	1.00	0.95	0.23	0.02	0.001

- › Almost every tree splits at the **root** (0.95), but a branch reaches **depth 3 only 2% of the time**
- › The prior concentrates on **shallow trees**

BART: POSTERIOR SIMULATION

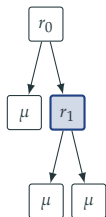
- **Backfitting**: Fit each tree to what the **other** trees leave unexplained
- That is, sample tree s on the **partial residual** $y_s^* = y - \sum_{j \neq s} g(x \mid \mathcal{T}_j, \mu_j)$, which carries all the information in the other $S - 1$ trees
- Update the tree by **Metropolis Hastings**, with one of four local moves (next slide); the leaf values are integrated out, so the tree can grow or shrink freely
- Given the trees, the leaves μ_s and σ^2 are simple Gibbs updates (see Session 2)

THE FOUR PROPOSAL MOVES

From the current tree, Metropolis Hastings proposes one **local move** (r : a splitting rule, μ : a leaf; changed node in blue):

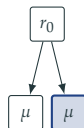
Grow

split a leaf



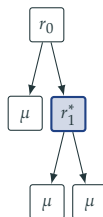
Prune

merge two leaves



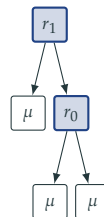
Change

new splitting rule



Swap

exchange two rules



BART IN MACROECONOMIC APPLICATIONS

- **Multivariate BART for US macro forecasting** (Clark, Huber, Koop, Marcellino & Pfarrhofer (2023, *IER*))
- **Nowcasting through the pandemic** (Huber, Koop, Onorante, Pfarrhofer & Schreiner 2023, *JoE*): Regression trees to model the extreme COVID-19 observations
- **Heteroskedastic BART** (Pratola, Chipman, George & McCulloch 2020, *JCGS*): A nonparametric alternative to stochastic volatility (SV)
- **Interpretable, semiparametric BART** (Hauzenberger, Huber, Koop & Mitchell forthcoming, *AoAS*; Deshpande et al. 2024, *BA*): Trees model the **time-varying coefficients**, keeping the mean conditionally linear and directly interpretable (see end of this session)

Gaussian process regression

GAUSSIAN PROCESS (GP) REGRESSION

- › Classic regression restricts f itself (linear, polynomial bases); smooth, but restrictive and ad hoc
- › A GP introduces **smoothness through the prior**, which is placed directly on the function
- › **No explicit assumptions** on the functional form:

$$f(x_t) \sim \mathcal{GP}(0, k_{\vartheta}(x_t, x_{\tau})) \quad \Rightarrow \quad f \sim \mathcal{N}(\mathbf{0}_T, \mathbf{K}), \quad [\mathbf{K}]_{t\tau} = k_{\vartheta}(x_t, x_{\tau})$$

with ϑ denoting a set of hyperparameters

- › For an observed sample, the infinite-dimensional GP prior collapses to just a finite T -dimensional Gaussian
- › k_{ϑ} measures **similarity**: When x_t and x_{τ} are close, $f(x_t)$ and $f(x_{\tau})$ are close

THE SQUARED-EXPONENTIAL (GAUSSIAN) KERNEL

- The functional characteristics are governed by the choice of kernel
- By far the most popular choice is the **squared-exponential (Gaussian)** kernel:

$$k_{\mathfrak{g}}(\mathbf{x}_t, \mathbf{x}_\tau) = \xi \times \exp\left(-\frac{\|\mathbf{x}_t - \mathbf{x}_\tau\|^2}{2\ell^2}\right), \quad \mathfrak{g} = \{\xi, \ell\}$$

with ξ the variance and ℓ the **length scale**

- The length-scale is the most important tuning choice:
 - **Large ℓ** : All points are similar $\Rightarrow$ smooth, near-linear f
 - **Small ℓ** : Only very close points are similar $\Rightarrow$ flexible f
- Matches many features of **(stationary) time series**

THE SE KERNEL IN NUMBERS: FIVE TIME POINTS

- Time as the input ($x_t = t, t = 1, \dots, 5; \xi = 1$)
- Kernel depends only on the lag $d = |t - \tau|$; comparison to an AR(1) with persistence 0.9

$$k(t, \tau) = \exp\left(-\frac{d^2}{2\ell^2}\right)$$

SE kernel, $\ell = 4$

$t \backslash \tau$	1	2	3	4	5
1	1.00	0.97	0.88	0.75	0.61
2	0.97	1.00	0.97	0.88	0.75
3	0.88	0.97	1.00	0.97	0.88
4	0.75	0.88	0.97	1.00	0.97
5	0.61	0.75	0.88	0.97	1.00

AR(1) kernel: 0.9^d

$t \backslash \tau$	1	2	3	4	5
1	1.00	0.90	0.81	0.73	0.66
2	0.90	1.00	0.90	0.81	0.73
3	0.81	0.90	1.00	0.90	0.81
4	0.73	0.81	0.90	1.00	0.90
5	0.66	0.73	0.81	0.90	1.00

GP POSTERIOR AND PREDICTION

- The GP posterior mean is a **kernel-weighted average of the observed y 's**; observations whose x are close count more (see next slide)
- Conditional posterior is available in **closed form**:

$$f \mid y, \sigma^2 \sim \mathcal{N}\left(K(K + \sigma^2 I_T)^{-1}y, \quad K - K(K + \sigma^2 I_T)^{-1}K\right)$$

- Predictions at a new x_{T+h} take the same form
- The posterior variance says **how much we learn**: $\sigma^2 \rightarrow \infty$ returns the prior (no learning)
- Conditional posterior of σ^2 has its usual inverse Gamma form

GP PREDICTION: A SMART WEIGHTED AVERAGE

- Point by point, that posterior mean is a **linear combination of the observed y 's**:

$$\hat{f}(x_*) = \sum_{\tau} w_{\tau} y_{\tau}$$

with $w' = K_{\vartheta}(x_*, X)(K + \sigma^2 I_T)^{-1}$

- Continuing the example ($\ell = 4, \sigma^2 = 1$), predict one-step ahead ($t_* = 6$):

observation τ	1	2	3	4	5
similarity $k(t_*, \tau)$	0.46	0.61	0.75	0.88	0.97
weight w_{τ}	-0.02	0.04	0.12	0.22	0.32

- The weights are that **similarity row** $K_{\vartheta}(x_*, X) \times (K + \sigma^2 I_T)^{-1}$

GP REGRESSION IN MACROECONOMICS

- GPs suit macro data well: $\mathbf{X}$ (typically huge K) **never enter directly**; only through the $T \times T$ kernel matrix $K_{\vartheta}(\mathbf{X}, \mathbf{X})$
- Inversion $(K_{\vartheta}(\mathbf{X}, \mathbf{X}) + \sigma^2 \mathbf{I}_T)^{-1}$ is computationally costly; fine for quarterly macro data (small T), prohibitive for daily financial data (large T)
- **GP-VARs** (Hauzenberger, Huber, Marcellino & Petz 2025, *JBES*) let the conditional mean be an unknown function of the lags, capturing asymmetric transmission and time-varying uncertainty
- **GP vs BART**: GP **smooths** where BART takes **piecewise-constant steps**

Bayesian neural networks (BNNs)

NEURAL NETWORKS (NNs)

- NNs can **approximate any nonlinear relation arbitrarily well** and have proved useful across many fields
- Their use in macroeconometrics has for long been limited
 - NNs are usually trained on **vast datasets** with **millions of parameters**
 - Macro datasets have a **few hundred observations** and almost as many candidate series
- Estimating fully-fledged multi-layer NNs on macro data is challenging

IMAGE RECOGNITION: PAIN AU CHOCOLAT OR SLOTH?



Running example for NNs, see, e.g., [kaggle](#)

- › **Binary classification:** Given images of sloths and pastries, sort each into one of two groups
- › **Inputs** are pixel values; only a **highly nonlinear** map from pixels to label separates them; far beyond any linear function
- › Training such a model can take **millions of labelled images**
- › **Thought experiment:** If the training data contain only pastries and never a sloth, what happens when a sloth appears? Flexible learners typically **extrapolate poorly**

A NEURAL NETWORK

- Goal: Estimate an unknown, possibly nonlinear relationship between a macro series $y_t \in \mathbb{R}$ and K covariates $\mathbf{x}_t \in \mathbb{R}^K$
- Start from the general nonlinear regression

$$y_t = f(\mathbf{x}_t) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

- An NN is a flexible functional form for the unknown f ; a **universal approximator** (Hornik et al. 1989, *Neural Networks*)
- We distinguish **shallow** (one hidden layer) and **deep** (several hidden layers) networks

A SHALLOW NEURAL NETWORK

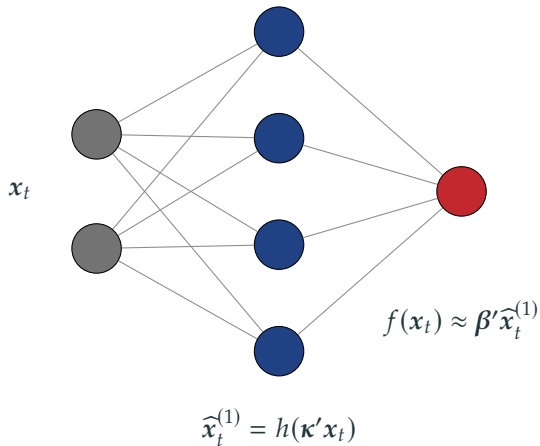
Shallow network = a single hidden layer:

$$f(\mathbf{x}_t) \approx \sum_{q=1}^Q \beta_q h_q(\mathbf{x}'_t \boldsymbol{\kappa}_q + \zeta_q)$$

- › $\boldsymbol{\beta} = (\beta_1, \dots, \beta_Q)'$: $Q \times 1$ vector of output **loadings**
- › Q : Number of **neurons**
- › $h_q(\bullet)$: Nonlinear **activation function** (e.g., $\text{ReLU}(x) = \max(0, x)$, sigmoid, tanh)
- › $\boldsymbol{\kappa}_q$: **Network weights** (collected in the $K \times Q$ matrix $\boldsymbol{\kappa}$)
- › ζ_q : **Bias** (intercept)

THE ARCHITECTURE OF A SHALLOW NEURAL NETWORK

Input layer Hidden layer Output layer



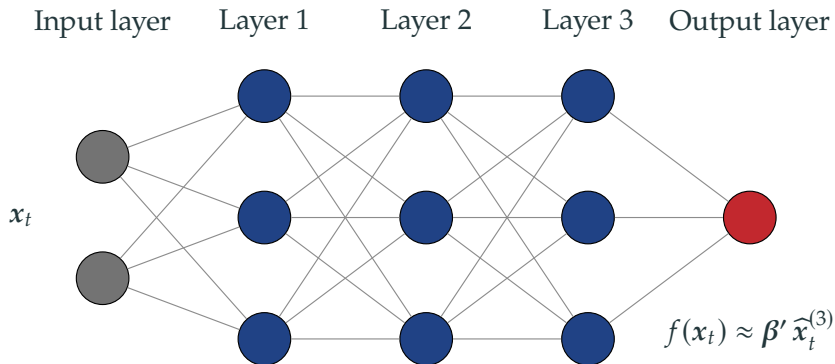
A DEEP NEURAL NETWORK

- ▶ Instead of a single layer, use L hidden layers with Q_ℓ neurons in layer ℓ ($\ell = 1, \dots, L$):

$$f(x_t) \approx \beta' h_L(W_L h_{L-1}(\cdots W_2 h_1(W_1 x_t)))$$

- ▶ W_ℓ is a $Q_\ell \times Q_{\ell-1}$ weight matrix ($\ell = 2, \dots, L$); W_1 is $Q_1 \times K$; β is $Q_L \times 1$ (biases ζ_ℓ suppressed)
- ▶ Setting $L = 1$ recovers the shallow network

THE ARCHITECTURE OF A DEEP NEURAL NETWORK



$$\hat{x}_t^{(1)} = h_1(W_1 x_t) \quad \hat{x}_t^{(2)} = h_2(W_2 \hat{x}_t^{(1)}) \quad \hat{x}_t^{(3)} = h_3(W_3 \hat{x}_t^{(2)})$$

A GENERAL NONLINEAR NN REGRESSION

- › Hauzenberger, Huber, Klieber & Marcellino (2025, *JoE*) use:

$$y_t = \underbrace{x_t' \gamma}_{\text{linear part}} + \underbrace{f(x_t)}_{\text{nonlinear part}} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2)$$

- › **Linear part** with a horseshoe prior on γ (global-local shrinkage of Session 2)
- › **Nonlinear part** f approximates remaining nonlinearities
- › In addition, **switch off inputs within a neuron** (horseshoe on κ_q)
- › σ_t^2 follows a **SV** process (Sessions 4); essential for macro data with volatility clustering and outliers (COVID-19, GFC)

Stylistic application: GDP growth on NFCI

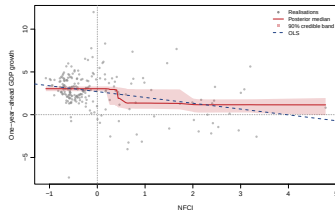
Single-covariate nonparametric GaR regression

GAR: A NONPARAMETRIC CONDITIONAL MEAN

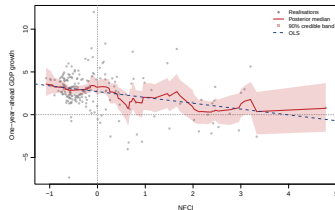
- **Recall previous sessions:** Regression of one-year ahead GDP growth on the NFCI
- **Data** (exactly as before): Annualised average US real GDP growth on a **single covariate** NFCI
- Learn the conditional mean $f(\text{NFCI}_t)$ **nonparametrically** (BART / GP / BNN) and let the data decide the functional form

GAR with BART: VARY THE NUMBER OF TREES S

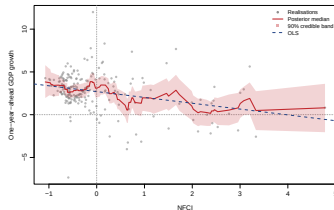
$S = 1$ tree



$S = 25$ trees



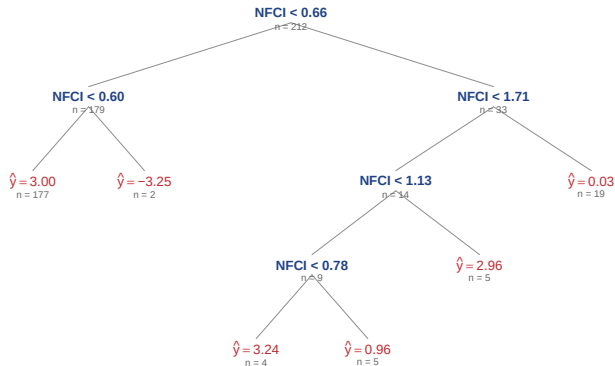
$S = 250$ trees



- ▶ $S = 1$ is a crude **step function**; adding trees **smooths** the fit; shrinkage keeps each tree a weak learner

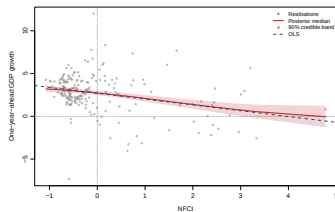
GAR with BART: INSIDE THE SINGLE TREE ($S = 1$)

One of the deeper posterior tree draws:

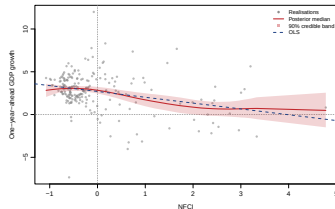


GAR with a GP: VARYING LENGTH-SCALE PARAMETER ℓ

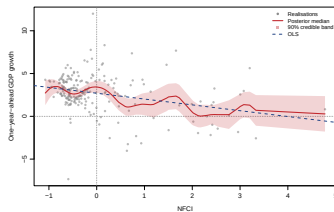
$\ell = 4$ (very smooth)



$\ell = 1$ (medium)

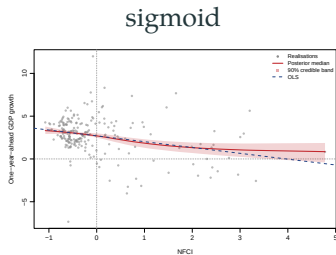
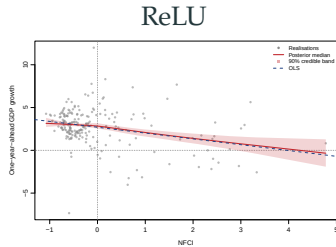
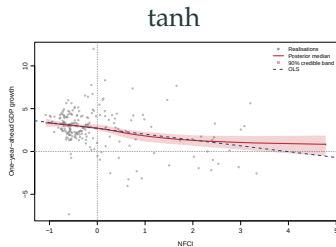


$\ell = 0.25$ (very flexible)



- $\ell = 4$ is **indistinguishable from OLS**, $\ell = 0.25$ **overfits**; in practice ℓ should be estimated or selected, e.g., via the marginal likelihood

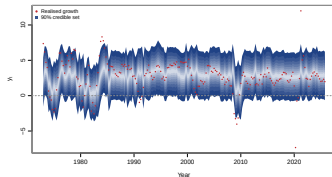
GAR with a shallow BNN: VARYING ACTIVATION FUNCTIONS



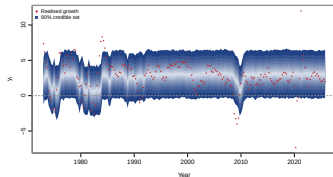
- Posterior median stays **close to OLS**, and the nonparametric part shows up mainly as **widening credible bands** once conditions tighten

GAR: IN-SAMPLE PREDICTIVE FIT, ONE PER MODEL CLASS

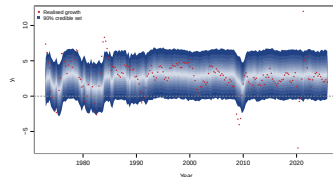
BART, $S = 250$



GP, $\ell = 1$



BNN, tanh



Interpretability and variable selection

VARIABLE SELECTION AND INTERPRETABILITY

- A common criticism of ML methods is that they are **black boxes**
 - ML reduce the risk of **misspecification** (+)
 - But the marginal effect of x_t on y_t is hard to identify (−)
- BART has built-in tools for interpretability
 - **Variable inclusion proportions (VIP)**: Proportion of splitting rules that use predictor j across all trees and MCMC draws
 - Acts as a Bayesian variable importance measure
 - Can be used to identify which predictors drive the nonlinear relationship

APPROXIMATE MARGINAL EFFECTS: LINEAR POSTERIOR SUMMARIES

- › **Effect-size analogues** (Crawford et al. 2018, *JASA*; Crawford et al. 2019, *AoAS*) project covariates X onto the fitted function f

$$\tilde{\beta} = \text{Proj}(X, f) = X^+ f$$

- › **In macro:** Huber, Koop, Onorante, Pfarrhofer & Schreiner (2023, *JoE*) use this projection in their nonparametric mixed-frequency VAR, both to run FFBS on the latent monthly states via $y_t = x_t' \tilde{\beta} + \varepsilon_t$ and as an interpretable, linear summary of the BART fit
- › **Surrogate models** (Woody, Carvalho & Murray 2021, *JCGS*): Project each posterior draw of f onto a simpler structure

A SEMIPARAMETRIC, INTERPRETABLE TREE MODEL

- A **semiparametric** model that keeps interpretability
- Recall the TVP model (Session 3), where f is **conditionally linear**: $f(x_t) = \beta_t x_t$
- Treat **coefficients (not variables)** nonparametrically
 - Approximate the TVPs with a tree, $\beta_t = f(z_t)$, driven by **effect modifiers** z_t (economic quantities that move y_t only **indirectly**, through β_t)
 - Mean relationship stays **conditionally linear**, so β_t is directly interpretable as a **marginal effect**
- Used in the paper I present in the research seminar